No Cones, No Duals: A Fast and Scalable Solver for NRS*f*M

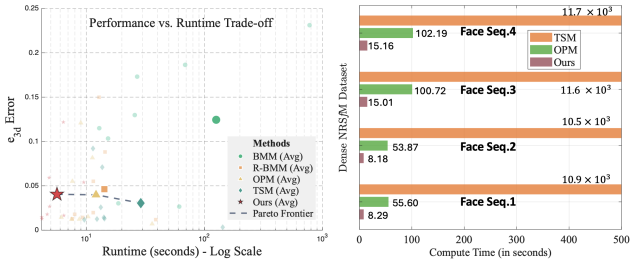
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Abstract

Non-rigid structure from motion (NRSfM) aims to recover camera motion and time-varying 3D shape from tracked 2D image measurements. Recent high-performing methods show that shape-space factorization remains highly effective for this task. These methods often use costly solvers for two key steps. First, a semidefinite programming (SDP) cone solver to perform Euclidean correction. Next, an alternating direction method of multipliers (ADMM) solver to recover the low-rank shape. These steps often scale poorly to long sequences and dense point tracks. We propose a fast, scalable NRSfM solver that avoids these costly steps. For rotation estimation, we introduce a differentiable Euclidean correction that directly optimizes small factors for each sequence. A row-polar projection then maps each estimated camera block to the closest row-orthonormal matrix. For shape estimation, we introduce an unrolled spectral-proximal solver. It combines reprojection-gradient steps with weighted singular-value shrinkage in the reshuffled shape domain. This solver further uses FISTA-style restart and continuation. The spectral shrinkage operator itself is classical, but we embed it in a fixed differentiable solver, while preserving NRSfM's geometric constraints. The proposed approach is category-agnostic, requires no external training data, and maps naturally to modern accelerator-friendly linear algebra primitives. Across sparse, dense, noisy, missing-data, and long-sequence benchmarks, our solver achieves substantial speedups with competitive reconstruction accuracy, including 7×–1000× runtime improvements on dense sequences with 28,887 tracked points.

1. Introduction

This paper concerns a matrix-factorization approach to solve Non-Rigid Structure from Motion (NRS*f*M). It is a long-standing inverse problem in geometric computer vision [30]. It seeks to recover camera motion and the 3D shape of a dynamically deforming object from tracked 2D image measurements. NRS*f*M applications include medical imaging [18, 31, 56], deformable and soft robotic perception [41], motion capture [5, 34, 36, 38], and visual effects [23, 51].



a) 3D reconstruction error-runtime Pareto comparison on CMU MoCap. Lower is better.

b) Compute Time Comparison on Dense Face Dataset with SOTA

Figure 1. Accuracy-runtime and dense-sequence behavior. (a) Accuracy-runtime Pareto comparison on CMU MoCap [5]. Smaller, faded markers show individual sequences, while larger markers show method averages (lower left is better). The proposed solver achieves the fastest runtime while retaining competitive reconstruction accuracy relative to recent shape-space baselines. (b) On dense sequences [18] with 28, 887 tracks, the proposed solver achieves large runtime gains over recent shape-space baselines such as TSM [57] and OPM [33], while retaining competitive accuracy.

NRS*f*M factorization remains difficult because many plausible motion-shape decompositions can often explain the same 2D tracks, and both the camera motion and the deforming 3D shape are unknown [32, 40]. Prior work also studies matrix factorization approaches under weak-perspective [20, 28, 62] and perspective camera models [14, 42]. This paper focuses on the classical orthographic setting with tracked image points. The measurement matrix is modeled as the product of a block-orthographic camera matrix and a shape matrix. Each camera block has orthonormal rows, while the deforming shape is assumed to span a low-rank space.

Factorized NRS*f*M representations include trajectory-space [6], shape-space [11], and shape-trajectory models [35]. Other approaches use force-based models [3], unions of subspaces [34, 35, 69], unions of piecewise curve subspaces [2], or probabilistic point-trajectory models [4]. Among these, trajectory-space and shape-space methods are particularly influential and widely used. Akhter et al. [5, 6]

pioneered the *trajectory-space* view. It represents each 3D point's motion with a small set of temporal basis functions. Naturally, the popular choice here is the discrete cosine transform (DCT) basis. This representation is fast to compute. However, a small fixed temporal basis may poorly capture complex or high-frequency deformations [53, 68]. This can limit reconstruction accuracy [26]. A dual view to this problem is shape-space approaches. Bregler et al. [11] introduced this in early 2000. Dai et al. [15] and Kumar and Van Gool [33] later advanced this approach and showed outstanding results. These methods model shape at each frame and impose a low-rank constraint across time. With careful Euclidean correction, they often achieve higher reconstruction accuracy than trajectory-space methods [26, 33]. Nevertheless, this accuracy comes at a computational price.

Recent shape-space methods [33, 57] achieve excellent reconstruction results but rely on two costly solver components. First, Euclidean correction estimates or camera rotation initialization. This step often uses semidefinite programming (SDP) or related global optimization over Gram matrices [15, 33, 57]. Second, a low-rank recovery step estimates shape with the rotations fixed. It typically uses nuclear or weighted nuclear penalties and ADMM-style updates [10, 33, 57]. SDP cone solvers used for rotation estimation are difficult to parallelize efficiently. Meanwhile, ADMM introduces auxiliary variables, dual updates, and large subproblem solves. Moreover, it is sensitive to the method's stopping criteria. These costs become a practical bottleneck as video grow longer or point tracks become denser.

We ask whether a fast and scalable solver can preserve the algebraic structure behind accurate shape-space NRS*f*M. Our answer is the **No Cones, No Duals** solver. No Cones means that rotation recovery avoids SDP cone solvers. It optimizes a small factor parameterization directly rather than solving for a positive semidefinite Gram matrix. No Duals means that shape recovery avoids ADMM dual variables and large shape-subproblem solves. It replaces alternating primal, auxiliary, and dual updates with a fixed unrolled spectral-proximal loop. The solver retains the algebraic and geometric structure of classical NRS*f*M. It expresses this structure through differentiable linear algebra operations suited to efficient computation on accelerators.

Rotation Solver. Classical Euclidean correction solves for Gram matrices satisfying positive semidefiniteness, rank, and orthonormality constraints [15]. We instead parameterize each triplet-side Gram matrix to enforce positive semidefiniteness and rank at most 3, while avoiding the explicit SDP cone solver. The factors are optimized using first-order gradient descent with the introduced rotation loss, which measures row-norm equality and row orthogonality after multiplication by the factorized motion matrix. The resulting camera blocks are then mapped to the closest row-orthonormal matrices by a row-polar projection. Thus, the factor-side parameterization, orthonormality loss, and row-polar projection together replace SDP-based Euclidean correction.

Stage	**Current Solvers**	**Ours**
Rotation	SDP/cone solver for Gram recovery	Factor-side $\mathbf{Y}_k$ + row-polar projection
Shape	ADMM with auxiliary/dual variables	Unrolled layer, WSVT shrinkage, restart
Camera	Orthographic camera constraints	row-polar camera projection
Computation	Cone, ADMM subproblem machinery	Accelerator-friendly solver

Table 1. No-cone, no-dual positioning relative to recent high-performing shape-space NRS*f*M factorization pipelines.

Shape Solver. We optimize the shape objective function via an ADMM-free spectral-proximal solver. We embed the spectral-proximal shrinkage operator in a fixed differentiable NRS*f*M solver layer that preserves the low-rank shape prior while removing ADMM's auxiliary and dual variables[1]. Each layer performs a reprojection-gradient step, reshapes the estimate into the standard shape-space matrix, applies weighted singular-value shrinkage, reshapes back, and uses FISTA-style restart and continuation for stable progress. This gives predictable runtime for a fixed number of layers (deterministic), while retaining the option to run more layers when accuracy is preferred over speed.

Our paper's positioning also differs from supervised deep learning-based NRS*f*M methods [17, 29, 49, 54, 58]. Supervised deep NRS*f*M methods learn category-level priors from training sequences and are therefore complementary to classical factorization. Since our goal is a category-agnostic per-sequence solver, we focus our comparison on factorization-based NRS*f*M methods, while reporting additional comparisons to learning-based approaches. Our solver remains category-agnostic and sequence-specific. It uses differentiable optimization modules, but it does not learn an external object prior. Meanwhile, it is also useful to distinguish our shape solver from dense variational NRS*f*M, especially Garg et al. [18], as it was an earlier attempt at an accelerator-friendly approach. Garg et al. [18] introduced a dense surface-reconstruction formulation that combines trace-norm regularization with edge-preserving total-variation spatial smoothness on an image-grid surface, and alternates rotation refinement with shape optimization through auxiliary variables, primal-dual TV updates, linear-system solves, and soft-impute-style shrinkage. On the contrary, our method estimates rotations through factor-side Euclidean correction and then applies a fixed no-dual shape solver. The solver, sequential approach pose and shape estimation, and the optimization procedure are different.

[1]We emphasize that weighted singular-value shrinkage and proximal-gradient methods are classical tools and not our claimed novelty.

Across sparse and dense benchmarks, our solver gives a better accuracy-runtime tradeoff. Note that it is not designed to dominate every specialized prior in 3D reconstruction error metric, but it consistently improves scalability while remaining competitive in accuracy. In particular, on dense sequences it provides large speedups over recent shape-space baselines while avoiding the SDP and ADMM machinery. Figure 1 and Table 1 summarize the proposed solver's performance and positioning, respectively. Overall, this paper makes the following contributions:

- We introduce an SDP-free factor-side Euclidean correction for NRS*f*M rotation estimation. The method optimizes small per-sequence factors and obtains valid orthographic camera blocks via row-polar projection, avoiding explicit cone solvers for Gram recovery.
- We introduce an ADMM-free unrolled spectral-proximal solver for low-rank shape recovery. Our approach removes ADMM dual variables and large shape-subproblem solves.
- We put forward a fast and scalable approach, which is category-agnostic, requires no external training data, and maps to accelerator-friendly linear algebra primitives.
- We evaluate the solver across sparse, dense, missing-data, noisy, and long-sequence benchmarks, showing substantial runtime gains with competitive reconstruction accuracy.

The central contribution is not a new low-rank prior or a new spectral shrinkage operator or the first scalable or gradient-based NRS*f*M method, it is a new solver pathway for shape-space NRS*f*M that replaces the prevailing conic and dual-variable solver machinery.

2. Preliminaries

Problem setup. We are given a mean-centered measurement matrix $\mathbf{W} \in \mathbb{R}^{2F\times P}$ containing the tracked 2D coordinates of P points across F frames [11, 60]. Each frame contributes two rows corresponding to the image x–y coordinates of all tracked points. Under the orthographic camera model, the standard NRS*f*M factorization writes $\mathbf{W} \approx \mathbf{RS}$, where $\mathbf{R} \in \mathbb{R}^{2F\times 3F}$ is block diagonal and $\mathbf{S} \in \mathbb{R}^{3F\times P}$ stacks the time-varying 3D shapes. The unknowns are constrained in two different ways. *(i)* each camera block must have orthonormal rows, i.e., $\mathbf{R}_i\mathbf{R}_i^T = \mathbf{I}_2$, and *(ii)* the deforming 3D shape must admit a low-dimensional representation over time [5, 13]. The difficulty of NRS*f*M comes from satisfying these constraints while explaining the same 2D observations through the bilinear product $\mathbf{RS}$.

• **Rotation Estimation.** A standard route to camera recovery begins with a rank-$3K$ factorization of the mean-centered measurement matrix,

$$\mathbf{W} \approx \hat{\Pi}\hat{\mathbf{B}}, \quad \text{where } \hat{\Pi} \in \mathbb{R}^{2F\times 3K}, \quad \hat{\mathbf{B}} \in \mathbb{R}^{3K\times P},$$

and K is the chosen shape-space rank. This factorization is not unique and for any invertible matrix $\Phi \in \mathbb{R}^{3K\times 3K}$,

$$\mathbf{W} = (\hat{\Pi}\Phi)(\Phi^{-1}\hat{\mathbf{B}}) := \Pi\mathbf{B}.$$

The role of Euclidean correction is to determine an upgrade Φ such that the upgraded motion matrix contains valid orthographic camera blocks. Let $\Phi = [\Phi_1, \Phi_2, \ldots, \Phi_K]$, where each triplet $\Phi_k \in \mathbb{R}^{3K\times 3}$ corresponds to one shape basis component. Classical shape-space methods introduce the triplet Gram matrix $\Psi_k := \Phi_k\Phi_k^T$. By construction, $\Psi_k \succeq 0$ and $\mathrm{rank}(\Psi_k) \le 3$, with rank three desired in the nondegenerate case. Let $\hat{\pi}_{2i-1}$ and $\hat{\pi}_{2i}$ denote the $(2i-1)$-th and $2i$-th rows of $\hat{\Pi}$, respectively. If the upgraded triplet agrees with a scaled camera block, then

$$\hat{\pi}_{2i-1:2i}\Phi_k = c_{ik}\mathbf{R}_i, \tag{1}$$

for some scalar coefficient c_{ik}. Since the two rows of $\mathbf{R}_i$ are orthonormal, the two rows of $\hat{\pi}_{2i-1:2i}\Phi_k$ must have equal squared norm and zero inner product. Written in terms of Ψ_k, this gives the standard Euclidean-correction constraints

$$\hat{\pi}_{2i-1}\Psi_k\hat{\pi}_{2i-1}^T = \hat{\pi}_{2i}\Psi_k\hat{\pi}_{2i}^T, \quad \hat{\pi}_{2i-1}\Psi_k\hat{\pi}_{2i}^T = 0. \tag{2}$$

Solving Eq. (2) together with $\Psi_k \succeq 0$ and a low-rank recovery condition gives a principled metric upgrade [13, 15]. However, this formulation operates over positive semidefinite matrix variables and typically requires an SDP or related cone-solver machinery [15, 57]. This is the rotation-side bottleneck that our approach addresses.

• **Shape Estimation.** Once the camera blocks $\{\mathbf{R}_i\}_{i=1}^F$ are known, the remaining problem is to estimate the time-varying 3D shape. Let $\mathbf{w}_{ij} \in \mathbb{R}^2$ be the observed 2D point j in frame i, and let $\mathbf{s}_{ij} \in \mathbb{R}^3$ be the corresponding 3D point in $\mathbf{S}_i$. The reprojection term alone is insufficient, so shape-space NRS*f*M imposes a low-rank prior on the sequence of shapes. It is common to use the reshuffled shape matrix $\mathbf{S}^\sharp = g(\mathbf{S}) \in \mathbb{R}^{F\times 3P}$, whose i-th row contains the flattened $3\times P$ shape of the i-th frame [13, 24, 33, 57]. When the deforming object is described by K basis shapes, $\mathrm{rank}(\mathbf{S}^\sharp) \le K$ provides a direct shape-space low-rank constraint. A schematic rank-regularized formulation is

$$\begin{aligned} &\min_{\mathbf{S}} \frac{1}{2}\sum_{i=1}^{F}\sum_{j=1}^{P}\|\mathbf{w}_{ij} - \mathbf{R}_i\mathbf{s}_{ij}\|_2^2 + \lambda\,\mathrm{rank}(\mathbf{S}^\sharp), \\ &\text{subject to : } \mathbf{S}^\sharp = g(\mathbf{S}) \end{aligned} \tag{3}$$

The exact rank penalty is nonconvex. Practical methods therefore replace it by nuclear, weighted nuclear, or related spectral penalties and solve a split optimization problem, often using ADMM-style auxiliary variables and dual updates [33, 57]. This ADMM machinery is accurate but introduces additional memory, large shape-subproblems, and stopping-criterion sensitivity, which our approach addresses.

3. Proposed Approach

The proposed solver has two stages, mirroring the two bottlenecks discussed in Sec. §2. First, we recover camera rotations without solving an SDP cone program. Second, given the rotations, we recover shape without ADMM dual variables or large shape-subproblem solves.

3.1. Rotation Estimation

The classical Euclidean-correction step estimates Gram matrices Ψ_k satisfying positive semidefiniteness, low-rank structure, and the frame-wise orthonormality constraints in Eq. (2). This is principled, but it leads to a cone-constrained optimization problem over Ψ_k. Our rotation estimation takes a different approach. We change the optimization variable. Instead of optimizing the Gram matrix directly, we optimize its factor. Let $\Theta = \{\mathbf{Y}_k\}_{k=1}^K, \mathbf{Y}_k \in \mathbb{R}^{3K\times 3}$, be the set of learnable per-sequence factor variables. We parameterize

$$\Psi_k(\Theta) = \mathbf{Y}_k\mathbf{Y}_k^T. \tag{4}$$

This guarantees $\Psi_k \succeq 0$ and $\mathrm{rank}(\Psi_k) \leq 3$ by construction. Full non-degeneracy is encouraged by the scale regularizer below and by the final row-polar projection. Most importantly, Eq.(4) avoids the need to form and solve the global cone system $\mathbf{A}\mathrm{vec}(\Psi_k) = \mathbf{0}$ associated with Eq.(2). The learnable variables are the small factor tensors $\mathbf{Y}_k$ themselves and no MLP or CNN architecture is introduced.

For each frame i and basis index k, we synthesize the two rows $\mathbf{g}_{1,ik} = \hat{\pi}_{2i-1}\mathbf{Y}_k \in \mathbb{R}^{1\times 3}$, $\mathbf{g}_{2,ik} = \hat{\pi}_{2i}\mathbf{Y}_k \in \mathbb{R}^{1\times 3}$. Stacking them gives $\mathbf{\Omega}_{ik} \triangleq [\mathbf{g}_{1,ik};\ \mathbf{g}_{2,ik}] \in \mathbb{R}^{2\times 3}$. If $\mathbf{Y}_k$ matches the true corrective triplet Φ_k up to the NRS*f*M scale coefficient, then $\mathbf{\Omega}_{ik} = c_{ik}\mathbf{R}_i$ due to Eq.(1). Hence the two rows of $\mathbf{\Omega}_{ik}$ should have equal squared norm and zero inner product. We encode these two Euclidean-correction conditions with the nonnegative orthonormality residual

$$\mathcal{L}_{\mathrm{ortho}}(\Theta) = \frac{1}{FK}\sum_{i=1}^{F}\sum_{k=1}^{K}\Big[\big(\|\mathbf{g}_{1,ik}\|_2^2 - \|\mathbf{g}_{2,ik}\|_2^2\big)^2 + \big(\mathbf{g}_{1,ik}\cdot\mathbf{g}_{2,ik}\big)^2\Big]. \tag{5}$$

This is the factor-side counterpart of the two linear constraints in Eq. (2). As a result, instead of enforcing the constraint on Ψ_k via a cone solver, we evaluate their residuals after multiplication by $\mathbf{Y}_k$. Still, the loss in Eq.(5) can admit a trivial solution $\mathbf{Y}_k = \mathbf{0}$. We therefore add a scale regularizer $s_{ik} = 0.5\left(\|\mathbf{g}_{1,ik}\|_2^2 + \|\mathbf{g}_{2,ik}\|_2^2\right)$, as **scale loss**:

$$\mathcal{L}_{\mathrm{scale}}(\Theta) = \frac{1}{K}\sum_{k=1}^{K}\left(\frac{1}{F}\sum_{i=1}^{F}s_{ik} - \tau_s\right)^2,\ \tau_s \approx 1. \tag{6}$$

This softly fixes the average row scale while preventing the degenerate condition. To control the scale of the factor variables and optimization stability, we also include a mild trace or Frobenius regularizer as **trace loss**:

$$\mathcal{L}_{\mathrm{trace}}(\Theta) = \frac{1}{K}\sum_{k=1}^{K}\mathrm{trace}(\mathbf{Y}_k\mathbf{Y}_k^T) = \frac{1}{K}\sum_{k=1}^{K}\|\mathbf{Y}_k\|_F^2 \tag{7}$$

Together, the scale and trace regularizers (Eq.(6) and Eq.(7)) discourage degenerate factors and stabilize optimization. The key observation is that evaluating residuals in the measurement space via $\mathbf{g}_{1,ik}$, $\mathbf{g}_{2,ik}$ is cheaper and numerically more stable. Consequently, the global null space system $A\mathrm{vec}(\Psi_k) = \mathbf{0}$ is unnecessary once we parameterize via $\mathbf{Y}_k$. Combining all the above loss functions, gives our overall loss function as

$$\begin{aligned}&\mathcal{L}_{\mathrm{rot}}(\Theta) = \mathcal{L}_{\mathrm{ortho}}(\Theta) + \lambda_s\mathcal{L}_{\mathrm{scale}}(\Theta) + \lambda_t\mathcal{L}_{\mathrm{trace}}(\Theta),\\ &\Theta^\star = \arg\min_{\Theta}\mathcal{L}_{\mathrm{rot}}(\Theta).\end{aligned} \tag{8}$$

All terms in Eq. (8) are evaluated through matrix multiplications involving $\hat{\Pi}$ and $\mathbf{Y}_k$, and are therefore simple to optimize with first-order gradient methods. After optimizing Θ, we select the basis triplet with the smallest average orthonormality residual $k^\star =$

$$\arg\min_{k}\frac{1}{F}\sum_{i=1}^{F}\Big[\big(\|\mathbf{g}_{1,ik}\|_2^2 - \|\mathbf{g}_{2,ik}\|_2^2\big)^2 + \big(\mathbf{g}_{1,ik}\cdot\mathbf{g}_{2,ik}\big)^2\Big] \tag{9}$$

The selection of $k^\star$ is a final discrete model-selection step after optimizing the factor variables. For each frame, the selected block $\mathbf{\Omega}_{ik^\star} = [\mathbf{g}_{1,ik^\star};\ \mathbf{g}_{2,ik^\star}] \in \mathbb{R}^{2\times 3}$ is then mapped to the closest row-orthonormal camera block using

$$\mathbf{R}_i = (\mathbf{\Omega}_{ik^\star}\mathbf{\Omega}_{ik^\star}^T)^{-1/2}\mathbf{\Omega}_{ik^\star}. \tag{10}$$

Note that the unknown scale $c_{ik^\star}$ cancels in this row-polar projection, and the result satisfies $\mathbf{R}_i\mathbf{R}_i^T = \mathbf{I}_2$ whenever $\mathbf{\Omega}_{ik^\star}$ has full row rank. We compute the 2×2 inverse square root using Newton-Schulz iterations [27, 55], which are lightweight and accelerator-friendly. The full motion matrix is finally assembled as $\mathbf{R} = \mathrm{blkdiag}(\mathbf{R}_1, \ldots, \mathbf{R}_F)$. A derivation showing Eq.(10) as the closest row-orthonormal projection is provided in the supplementary Sec.§10.

3.2. Shape Estimation

Given $\mathbf{W}$ and recovered $\mathbf{R}$, we estimate $\mathbf{S}$. We start from the standard re-projection term with low-rank shape objective

$$\min_{\mathbf{S}}\frac{1}{2}\|\mathbf{M}\odot(\mathbf{W} - \mathbf{RS})\|_F^2 + \|\mathbf{S}^\sharp\|_{*,w};\ \text{s.t. } \mathbf{S}^\sharp = g(\mathbf{S}). \tag{11}$$

Here, $\mathbf{M} \in \{0,1\}^{2F\times P}$ is the visibility mask, where $\mathbf{M}_{ij} = 0$ indicates a missing observation, and $g(\cdot)$ reshapes $\mathbf{S} \in \mathbb{R}^{3F\times P}$ into $\mathbf{S}^\sharp \in \mathbb{R}^{F\times 3P}$, whose i-th row is the flattened $3\times P$ shape of frame i. The weighted spectral penalty is $\|\mathbf{S}^\sharp\|_{*,w} = \sum_i w_i\sigma_i(\mathbf{S}^\sharp)$, where the singular

values are sorted in nonincreasing order. Following prior weighted-nuclear formulations in NRS*f*M [32], the weights are computed from the singular values of an initial shape $\mathbf{S}^0 = \mathrm{pinv}(\mathbf{R})\mathbf{W}$. The objective in Eq. (11) has a smooth re-projection term and a non-smooth spectral regularizer. More abstractly, we write

$$J(\mathbf{S}) = f(\mathbf{S}) + \mathcal{G}(\mathbf{S}). \tag{12}$$

where, $f(\mathbf{S}) = 0.5\|\mathbf{M}\odot(\mathbf{W}-\mathbf{R}\mathbf{S})\|_F^2$ is a smooth function while $\mathcal{G}(\mathbf{S}) = \|\mathbf{S}^\sharp\|_{*,w}$ is non-smooth function. The gradient of the smooth term is

$$\nabla f(\mathbf{S}) = \mathbf{R}^T\big(\mathbf{M}\odot(\mathbf{R}\mathbf{S}-\mathbf{W})\big) \in \mathbb{R}^{3F\times P}. \tag{13}$$

Now, for $f(\mathbf{S})$ gradient step to be valid and stable, $\nabla f(\mathbf{S})$ must be *Lipschitz continuous*. Because the mask is nonexpansive and $\mathbf{R}$ is block-orthographic, Lipschitz condition will tell us that ∇f does not change too abruptly, and geometrically the curvature of f is bounded. Under orthographic projection, the gradient is Lipschitz with constant $L \le \|\mathbf{R}\|_2^2 \le 1$ [8, 44, 48], so $L \le 1$. A fixed step size $\alpha \in (0, 1/L]$ or $\alpha \in (0, 1]$ is valid. This provides a valid fixed step size for the smooth reprojection step. Since the weighted spectral penalty is generally nonconvex, we use monotone restart and continuation for stable practical descent. Refer to the supplementary material Sec.§12 for detailed proof and explanations on Lipschitz condition.

***(i)* Unrolled Spectral-Proximal Solver.** $\mathcal{G}(\mathbf{S})$ in Eq.(12) is a weighted nuclear norm (WNN) term and is non-smooth. Its derivative is undefined at zero singular values. To handle such a non-smooth regularizer, we used proximal operators, which are a standard tool in optimization [52]. It provides a closed-form spectral shrinkage step that can be inserted directly after the reprojection-gradient update. Thus, a proximal step is the building block that replaces the dual update in ADMM, but in a more direct and accelerator-friendly way.

Given $\mathbf{S}^\sharp \in \mathbb{R}^{F\times 3P}$, with SVD, $\mathbf{S}^\sharp = \mathbf{U}\,\mathrm{diag}(\{\sigma_i\})\mathbf{V}^T$, where σ_i are the singular values sorted in nonincreasing order. The weighted singular value thresholding (WSVT) operator acts by shrinking each singular value by an amount proportional to its weight w_i:

$$\mathcal{S}_{\tau,w}(\mathbf{S}^\sharp) = \mathbf{U}\,\mathrm{diag}\left(\{\max(\sigma_i - \tau w_i, 0)\}_i\right)\mathbf{V}^T. \tag{14}$$

That is, large singular values (important deformation modes) survive the shrinkage, while small ones, i.e., spurious deformations or noise, are driven to zero. Following [33], we use Eq.(14) ordered weighted shrinkage update. Because the weighted spectral penalty is generally nonconvex for the weights used here, we do not claim global convex optimality of the full objective. We use this operation as a closed-form spectral shrinkage update inside a monotone-restarted fixed-depth unrolled feed-forward solver [47, 64]. The regularizer remains non-smooth; yet, the shrinkage map is differentiable almost everywhere and can be embedded in an autograd-compatible computation graph.

We combine the reprojection-gradient step and the spectral shrinkage update as follows: (i) a gradient step on $f(\mathbf{S})$, and (ii) a weighted spectral-shrinkage step on $\mathcal{G}(\mathbf{S})$. Let $\mathbf{S}^{(k)}$ be the current shape iterate, and $\mathbf{T}^{(k)}$ the FISTA auxiliary variable [9, 64]. Each iteration performs: *1. Gradient Step*: $\mathbf{Z}^{(k)\sharp} = g(\mathbf{T}^{(k)}) - \alpha\, g(\nabla f(\mathbf{T}^{(k)}))$, *2. Proximal shrinkage and reshape*: $\mathbf{X}^{(k)\sharp} = \mathcal{S}_{\tau_k,w}(\mathbf{Z}^{(k)\sharp})$, $\mathbf{S}^{(k+1)} = g^{-1}(\mathbf{X}^{(k)\sharp})$, and *3. FISTA momentum [9, 64]:* $t_{k+1} = \frac{1}{2}\big(1+\sqrt{1+4t_k^2}\big)$; $\beta_k = \frac{t_k-1}{t_{k+1}}$, $\mathbf{T}^{(k+1)} = \mathbf{S}^{(k+1)} + \beta_k(\mathbf{S}^{(k+1)} - \mathbf{S}^{(k)})$. Here, α is a step size and $g(\nabla f(\cdot))$ reshapes the gradient into the reshuffled shape $\mathbf{S}^\sharp$ dimension. We use a monotone restart if the cost of $J(\mathbf{S})$ increases, i.e., if $J(\mathbf{S}^{(k+1)}) > J(\mathbf{S}^{(k)})$, we set $t_{k+1} = 1$ and $\mathbf{T}^{(k+1)} = \mathbf{S}^{(k+1)}$. Stacking a fixed number N_k of these iterations gives a deterministic feed-forward unrolled shape solver with predictable runtime. A detailed computational pipeline is provided in the supplementary material Sec. §9.

***(ii)* Continuation schedule and practical safeguards.** We set the shrinkage thresholds as $\tau_k = \mu/\rho_k$ (where $\mu \approx 1$) with $\rho_{k+1} = \min(\rho_{\max}, \eta\rho_k)$. Early iterations (small ρ) apply strong shrinkage that rapidly enforces low rank, while later iterations (larger ρ) relax the shrinkage to fine-tune reprojection accuracy. This plays a role analogous to ADMM penalty continuation, but without dual variables. We preserve the first N_{pres} singular values by setting the shrinkage amount $\Delta_i = 0$ for $i \le N_{\mathrm{pres}}$, ensuring that the dominant deformation mode is not shrunk. We also cap shrinkage by replacing $\tau_k w_i$ with $\min(\tau_k w_i, \mathrm{cap}\cdot\sigma_i(\mathbf{Z}^{(k)\sharp}))$ to prevent premature collapse of small but meaningful deformation modes. All continuation and safeguard parameters are fixed across experiments unless otherwise stated.

***(iii)* No-dual solver interpretation.** The shape stage replaces the ADMM split variables and dual updates with a direct feed-forward sequence layer of gradient and spectral-shrinkage operations. Each layer uses only matrix multiplication, masking, reshaping, SVD, scalar shrinkage, and a FISTA-style skip/restart [9, 64]. Thus, the method preserves the shape's low-rank without ADMM dual variables, large shape-subproblem linear solves, and dependence on the stopping criterion. A complexity comparison with ADMM-based solvers is provided in the supplementary Sec. §11.

4. Experiment and Results

***(i)* Implementation Details.** We implemented our approach using Python 3.12.11 and PyTorch 2.5.1. We report wall-clock runtimes on an Apple M3 Max unless otherwise noted. Dense sequence experiments requiring CUDA SVD were run on an NVIDIA RTX 6000 Ada GPU. Runtimes are indicative and emphasize scaling trends. When exact same-hardware reruns were not possible due to released-code or backend

Algorithm 1 Shape Solver

Require: $\mathbf{R}, \mathbf{W}, \mathbf{M}$; step size $\alpha \in (0, 1/L]$; shrinkage scale μ; weights w (non-decreasing); schedule parameters $\rho_0 > 0, \eta > 1, \rho_{\max}$; unrolling depth N_k; options: $N_{\text{pres}} = 1$, cap
Ensure: $\mathbf{S} \in \mathbb{R}^{3F \times P}$

Initialize $\mathbf{S}^{(0)} = \mathbf{R}^{\dagger}\mathbf{W}$, $\mathbf{T}^{(0)} = \mathbf{S}^{(0)}$, $t_0 = 1$, $\rho = \rho_0$, $\mu = 0.99$, cap $= 0.99$, $\eta = 1.05$, $N_k = 160$
for $k = 0$ to $N_k - 1$ **do**
  $\mathbf{G} \leftarrow \mathbf{R}^T(\mathbf{M} \odot (\mathbf{R}\mathbf{T}^{(k)} - \mathbf{W}))$ # gradient in original domain
  $\mathbf{Z}^{\sharp} \leftarrow g(\mathbf{T}^{(k)}) - \alpha g(\mathbf{G})$ # S-sharp gradient step
  $[\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}] \leftarrow \text{svd}(\mathbf{Z}^{\sharp})$; $\tau_k \leftarrow \mu/\rho_k$
  for $i = 1$ to $\min(F, 3P)$ **do** # weighted shrinkage
    $w_i \leftarrow \mathbf{w}(i)$; $\Delta_i \leftarrow \tau_k w_i$;
    if $i \leq N_{\text{pres}}$ **then** $\Delta_i \leftarrow 0$
    end if
    if cap > 0 **then** $\Delta_i \leftarrow \min\{\Delta_i,\ \text{cap} \cdot \mathbf{\Sigma}_{ii}\}$
    end if
    $\mathbf{\Sigma}'_{ii} \leftarrow \max\{\mathbf{\Sigma}_{ii} - \Delta_i,\ 0\}$
  end for
  $\mathbf{X}^{\sharp} \leftarrow \mathbf{U}\mathbf{\Sigma}'\mathbf{V}^T$; $\mathbf{S}^{(k+1)} \leftarrow g^{-1}(\mathbf{X}^{\sharp})$
  # FISTA with monotone restart
  $t_{k+1} = \frac{1}{2}(1 + \sqrt{1 + 4t_k^2})$; $\beta_k = \frac{t_k - 1}{t_{k+1}}$
  $\mathbf{T}^{(k+1)} = \mathbf{S}^{(k+1)} + \beta_k(\mathbf{S}^{(k+1)} - \mathbf{S}^{(k)})$
  # restart momentum if objective increases
  if $J(\mathbf{S}^{(k+1)}) > J(\mathbf{S}^{(k)})$ **then**
    Restart: $t_{k+1} = 1$, $\mathbf{T}^{(k+1)} = \mathbf{S}^{(k+1)}$
  end if
  $\rho_{k+1} \leftarrow \min(\rho_{\max}, \eta\rho_k)$ # continuation
end for
return $\mathbf{S}^{(N_k)}$

limitations, we report the available baseline runtimes and interpret them primarily as scaling indicators. Unless otherwise stated, Table 2 reports rotation-stage runtime, Table 3 reports shape-stage runtime after rotations are provided, and Table 5 reports dense shape-space solver runtime. A stage-wise runtime breakdown is provided in the supplement.

We ran 3,500 steps to minimize the loss function proposed in Eq.(8) with learning rate 10^{-3} and $\lambda_s = 1e^{-4}$, $\lambda_t = 1e^{-6}$ to recover the rotation $\mathbf{R} \in \mathbb{R}^{2F \times 3F}$. We ran $N_k = 160$ fixed unrolling steps of proximal gradient descent (PGD) to recover shape $\mathbf{S}^{3F \times P}$ by optimizing Eq.(11) using the pseudo code outlined in **Algorithm** 1. Note $N_k = 160$ is a conservative fixed budget; in most sequences the objective stabilizes earlier. We initialized $\eta = 1.05$, $\rho_0 = 2e^{-3}$, $\rho_{\max} = 1e^{3}$, $N_{\text{pres}} = 1$, and the weight w following [33] work. Refer to **Algorithm** 1 for step-by-step implementation details. Note that the variable cap $= 0.99$ prevents premature spectral collapse over iterative steps.

***(ii)* Evaluation Metric.** Evaluations are done on popular motion capture (MoCap) benchmarks [5, 62], NRS*f*M challenge [26], dense NRS*f*M dataset from Garg et al. [18], and recent H3WB [70] dataset. We used mean normalized 3D reconstruction error metric $e_{3d} = \frac{1}{F}\sum_{i=1}^{F} \|\mathbf{S}_i^{\text{est}} - \mathbf{S}_i^{\text{gt}}\|_F / \|\mathbf{S}_i^{\text{gt}}\|_F$ to report 3D reconstruction accuracy. Here, $\mathbf{S}_i^{\text{est}}$ and $\mathbf{S}_i^{\text{gt}}$ denote the estimated and ground-truth 3D shape for frame i, respectively. For experiments on NRS*f*M challenge benchmark [26], we adopt their official evaluation code. The e_{3d} is reported after registering the estimated shape to the ground truth. To assess pose recovery, we report the mean rotation error, defined as $e_R = \frac{1}{F}\sum_{i=1}^{F} \|\mathbf{R}_i^{\text{gt}} - \mathbf{R}_i^{\text{est}}\|_F$, where, $\mathbf{R}_i^{\text{est}}$ and $\mathbf{R}_i^{\text{gt}}$ represent the estimated and ground-truth rotation matrices for frame i, respectively.

Data	(F, P, K)	BMM [15]	R-BMM [32]	OPM [33]	TSM [57] ($\mathbf{R}_{\text{init}}$/ $\mathbf{R}_{\text{ref}}$)	Ours
Drink	(1102, 41)	0.0086	0.0240	0.0096	0.0086/ 1.3950	**0.0060**
	$K = 11$	(10.01)	(108.91)	(143.54)	(88.94)	**(8.51)**
Pickup	(357, 41)	0.1210	0.1271	0.1144	0.1210/ 1.3197	**0.1082**
	$K = 12$	(09.68)	(96.24)	(98.87)	(14.79)	**(7.79)**
Yoga	(307, 41)	0.0945	0.0944	0.0972	0.0945/ 0.1114	0.0996
	$K = 11$	(08.08)	(70.08)	(76.88)	(10.48)	**(7.73)**
Stretch	(370, 41)	0.0682	0.0682	0.0688	0.0682/ 1.4116	**0.0647**
	$K = 13$	(08.45)	(73.32)	(115.06)	(19.40)	**(7.83)**

Table 2. e_R and rotation-stage runtime comparison with baselines. All methods are re-simulated for given K. TSM uses two rotation estimates, i.e., BMM initialization $\mathbf{R}_{\text{init}}$ and the refined rotation $\mathbf{R}_{\text{ref}}$ after its refinement module. Our approach is considerably faster while achieving competitive or better rotation accuracy. The number in parentheses shows compute time in seconds.

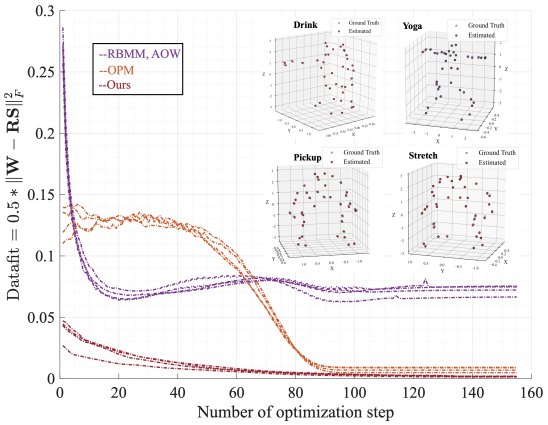


Figure 2. Stable convergence of the proposed solver on CMU MoCap sequences. The data-fit values $f(\mathbf{S})$ for R-BMM [32], AOW [24], and OPM [33] are normalized for visualization. Qualitative 3D reconstruction results obtained using our solver are also shown for representative sequences. The 4 curves correspond to data-fit values on 4 standard CMU sequences.

4.1. Dataset and Results

***(i)* MoCap Benchmark.** This benchmark introduced in [5, 62] is now a default evaluation suite for NRS*f*M algorithms. It comprises eight standard sequences: Drink (1102, 41), Pickup (357, 41), Yoga (307, 41), Stretch (370, 41), Dance (264, 75), Walking (260, 55), Face (316, 40), and Shark (240, 91). The first four sequences have both ground-truth (GT) rotation and 3D shape, whereas the remaining sequence have only GT 3D shape for evaluation. The last three sequences were introduced in [62]. The tuples augmented with each dataset represent the number of frames and points, i.e., (F, P). Shark is synthetic, while the remaining follow the MoCap protocol used in prior NRS*f*M evaluations.

(a) Rotation Estimation. Table 2 reports our e_R with compute time for comparison with other competing SOTA methods. Table 2 shows that our method attains accuracy better or close to SOTA shape-space based methods yet demonstrating a marked advantage in computational efficiency. For

Data	BMM [15]	R-BMM [32]	OPM [33]	TSM [57]	Ours
Drink	0.0266 (61.22)	0.0119 (38.75)	0.0071 (36.08)	0.0033 (142.96)	0.0166 **(06.45)**
Pickup	0.1731 (26.95)	0.0198 (11.44)	0.0152 (10.34)	0.0126 (12.31)	0.0267 **(04.88)**
Yoga	0.1150 (12.87)	0.0129 (07.60)	0.0122 (07.51)	0.0118 (09.62)	0.0148 **(04.19)**
Stretch	0.1034 (15.28)	0.0144 (09.32)	0.0124 (08.73)	0.0131 (14.24)	0.0128 **(04.21)**
Dance	0.1864 (68.68)	0.1499 (12.82)	0.1209 (8.98)	0.0921 (11.39)	0.1218 **(6.39)**
Face	0.0303 (18.68)	0.0179 (7.97)	0.0145 (6.140)	0.0144 (14.09)	0.0179 **(4.66)**
Walking	0.1298 (25.72)	0.0882 (14.40)	0.0816 (11.26)	0.0710 (13.64)	**0.0592** **(4.84)**
Shark	0.2311 (777.56)	0.0551 (11.46)	0.0551 (7.62)	0.0258 (13.26)	0.0540 **(9.51)**

Table 3. e_{3d} and shape-stage runtime comparison with high-performance methods after rotations are fixed. Best K from the previous methods is used to compare e_{3d} and time. Our approach is faster and its e_{3d} compares favorably with competing methods. Numbers in parentheses denote computation time in seconds.

instance, on the Drink sequence, which has 1102 frames and is nearly 3× longer than the other sequences in Table 2, the runtime of OPM [33] and TSM [57] increases substantially, whereas our solver exhibits only a modest increase in compute time, highlighting its scalability to longer sequences. For sequences without GT rotations, the supplement further reports row-orthonormality residuals $\|\mathbf{R}_i\mathbf{R}_i^T - \mathbf{I}_2\|_F$ (refer Figure 7) and a rotation-transfer experiment in which our recovered rotations are used inside existing shape solvers (refer Table 12).

(b) Shape Estimation. Table 3 shows the intended accuracy-runtime tradeoff. In addition to the favorable performance, our solver is consistently faster and remains competitive across the benchmark, supporting the goal of a fast no-cone, no-dual shape-space solver. For instance, on the drink sequence that has more than 1000 frames, our solver provides results in significantly less compute time.

Figure 2 provides qualitative results as well as convergence behaviors in comparison with recent ADMM based methods such as RBMM [32], AOW [24], and OPM [33]. The curves show data-fit values ($f(\mathbf{S})$) over iterations for different sequences, demonstrating stable convergence of our solver. Additionally, the plot demonstrates our approach robustness and computational efficiency relative to high-performing methods. *For more results and comparison with trajectory space methods refer supplementary material.*

***(ii)* NRS*f*M CVPR 2017 Benchmark.** This is a comprehensive benchmark dataset for evaluating NRS*f*M algorithms proposed by Jensen et al. [26]. It consists of 5 distinct deformable subjects, namely, Articulated, Paper, Balloon, Stretch, and Tearing. Each subject exhibits a wide range of non-rigid deformations. The deformations are recorded under 6 camera trajectories, namely, circle, flyby, line, semi-circle, tricky, and zigzag, making this dataset exceptionally challenging with a diverse set of motion and viewpoint variations. For assessment, the dataset contains ground-truth 3D shape for one reference frame only per subject category.

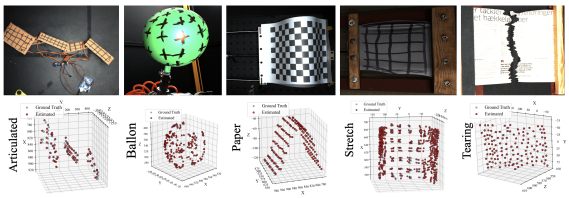


Figure 3. Qualitative results on NRS*f*M challenge dataset.

Data	CSF2 [22]	BMM [15]	R-BMM [32]	AOW [24]	BP [50]	OPM [33]	TSM [57]	Ours
Articulated	11.52	18.49	16.00	15.03	16.10	12.18	10.69	**10.58**
Balloon	10.14	10.39	7.84	8.05	8.29	6.29	7.28	**5.86**
Paper	9.72	8.94	10.69	10.45	**6.70**	8.86	7.91	10.42
Stretch	8.65	10.02	7.53	9.01	7.66	6.36	**5.43**	7.02
Tearing	12.04	14.23	16.34	16.20	11.26	10.91	10.77	**10.61**

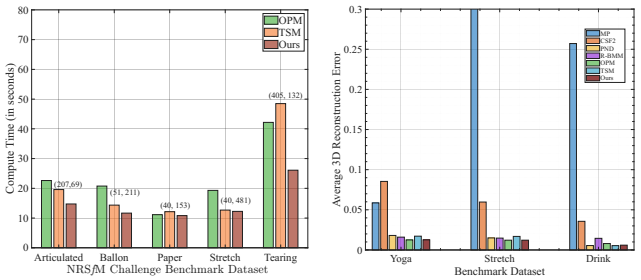
Table 4. 3D reconstruction error comparison with state-of-the-art on [26] dataset.

Data	SMSR [7]	DS [16]	GM [37]	ND [58]	DV [18]	OPM [33]	TSM [57]	Ours
Seq.1	0.1893	0.0636	0.0443	-	0.0531	0.0624 (55.60)	0.2223 (10964.22)	0.0774 **(8.29)**
Seq.2	0.2133	0.0569	0.0381	-	0.0457	0.0451 (53.87)	0.1271 (10500.35)	0.0442 **(8.18)**
Seq.3	0.1345	0.0374	0.0294	0.0450 (3h+ train)	0.0346	0.0279 (100.72)	0.0393 (11624.89)	0.0382 **(15.01)**
Seq.4	0.0984	0.0428	0.0309	0.0490 (3h+ train)	0.0379	0.0419 (102.19)	0.0615 (11609.17)	0.0522 **(15.16)**

Table 5. e_{3d} and overall run-time comparison with high-performing approaches on dense sequences [18]. Timings for other methods are unavailable due to the lack of open-source code. **Observation:** Our solver is faster than baselines: OPM and TSM.

Table 4 presents 3D reconstruction error (root mean square error (RMSE) in millimeters [26, 59]) averaged over 6 camera trajectory types. All evaluations are performed on the orthographic sequence of the dataset. The comparison includes recent as well as classical SOTA methods. Our solver is best or near-best on Articulated, Balloon, and Tearing, and favorably accurate on Paper and Stretch. These weaker cases involve challenging deformation and camera configurations where methods with stronger trajectory, temporal, or deformation priors can be advantageous. The main benefit remains favorable runtime and robustness across categories (refer Figure 4a). Figure 3 presents qualitative results, showing the reconstructed 3D shapes alongside the ground-truth 3D shapes. Overall, these results demonstrate the solver's speed, stability, and reconstruction quality under complex deformations and varying camera trajectories.

***(iii)* Dense NRS*f*M Benchmark.** Introduced by Garg et al. [18, 19], this dataset comprises of four synthetic face sequences with 28,887 tracked feature points. Face Seq.1 and Face Seq.2 are 10-frame-long whereas Face Seq.3 and Face Seq.4 are 99-frame long. Table 5 provides the statistical results of our approach compared to SOTA dense NRS*f*M methods. We use $K = 1$ to evaluate our approach on all the four sequences. Dense methods exploiting image-grid or spatial smoothness or latent space constraint can be better on some sequences [19, 37, 58]. Our solver does not use such priors and instead targets a general dense and sparse tracked-point setting with small as well as long sequences, yielding major runtime gains. Our computation



(a) Compute time comparison (b) Results on missing data

Figure 4. (a) Overall compute time comparison with current SOTA approaches [33, 57] on [26] dataset. **Observation**: Our approach compute time is significantly better as the number of frames and points increases (for e.g., Tearing seq. with $F = 405$, $P = 132$). (b) Our approach compares favorably with other methods on missing dataset sequence.

Data	CSF2 [22]	BMM [15]	PND [42]	CNS [43]	R-BMM [32]	TSM [57]	Ours
Eating2	0.0923	0.0665	0.0701	0.0823	0.0633	**0.0520**	0.0651
Smoking1	0.1230	0.0844	0.1049	0.1073	0.0746	0.0738	**0.0726**
Directions	0.0787	0.0648	0.0536	0.0621	0.0595	**0.0529**	0.0643
Smoking	0.0702	0.0492	0.0622	0.0578	0.0363	0.0352	**0.0337**
Waiting2	0.1029	0.1081	0.1114	0.1142	0.1117	0.1035	**0.1012**

Table 6. e_{3d} on One-circle-type camera motion dataset [70].

time is 7×–1000× faster than the reported OPM [33] and TSM [57] factorization baselines, demonstrating the scalability of the proposed solver on dense tracks. Note: For TSM, the default initialization for dense sequence leads to code break-down so we used CSF2 [22] initialization given in their code. Qualitative result on real dense sequence is provided in the supplementary

***(iv)* H3WB Dataset.** This dataset [70] extends Human3.6M [25] with 133 annotated keypoints across the full body and includes diverse daily actions. [57] created a new NRS*f*M evaluation benchmark using different camera motion types. Table 6 shows the e_{3d} comparison on the one-circle camera-motion setting. Our solver obtains the best result on several actions and remains competitive on the rest. Additional H3WB results are provided in the supplementary material.

4.2. Ablations

***(i)* Relevance of each term in rotation loss function.** We performed this ablation on CMU MoCap dataset [5] to demonstrate the utility of each term included in the rotation loss function—refer Eq.(8). Our rotation loss is composed of three main loss terms 1) Orthonormality loss 2) Scale loss, and 3) Trace loss. Figure 5 shows the change in $e_{\mathbf{R}}$ values as we add the proposed loss terms thereby demonstrating the usefulness of each term included in the proposed loss function for rotation estimation.

***(ii)* Performance on missing and noisy data sequence.** For missing trajectory sequence, we used setup as detailed in [12, 33], i.e., randomly set 30% of the trajectory missing from the $\mathbf{W}$. The results are shown in Figure 4b. Our method performs comparably with the SOTA approaches. For noisy trajectory sequence we again followed the experimental setup as described in [33]. We executed different

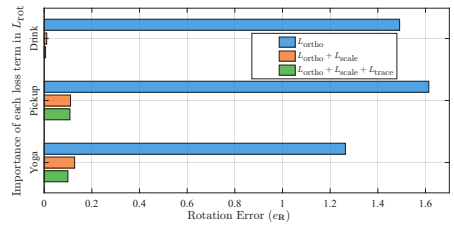
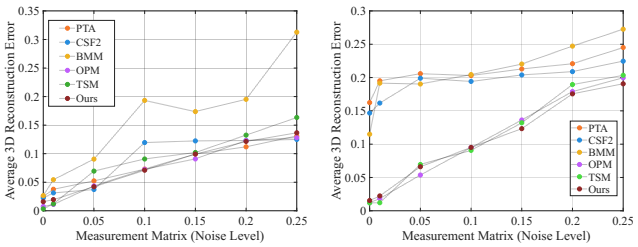


Figure 5. Importance of each loss term included in Eq.(8). Variation in $e_{\mathbf{R}}$ as we add the three loss terms.



(a) e_{3d} on noisy Drink seq. (b) e_{3d} on noisy Yoga seq.

Figure 6. e_{3d} variation w.r.t different noise levels in $\mathbf{W}$.

methods 10 times for each standard deviation value and plotted the method's mean. Figure 6a and Figure 6b show the comparison of different methods for e_{3d} on the noisy trajectory. Statistical analysis shows that our approach is robust to noise without noise-specific tuning and remains stable compared with SOTA approaches. *Refer to the supplementary material for more ablation.*

5. Limitations and Closing Remarks

Limitations. The proposed solver operates in the classical orthographic, tracked-point NRS*f*M setting. It assumes reliable 2D tracks, known visibility masks for missing data, and mean-centered measurements. Its shape stage approach does not use image-grid smoothness [18, 38, 45], explicit camera-trajectory or temporal deformation priors [39, 53], or category-level training [17, 46, 49]. This makes the solver broadly applicable, but specialized methods can be more accurate when their additional priors match the data, e.g., smooth dense surfaces or challenging camera motion. Extending our approach to perspective cameras and lightweight spatial or temporal priors is a natural future direction.

Closing Remarks. This work shows that shape-space NRS*f*M factorization need not depend on SDP cone solvers or ADMM dual variables for accurate rotation and shape recovery. It is shown that we can retain the well-founded algebraic principle of NRS*f*M factorization while making its computations simple, scalable, and accelerator-friendly by replacing Gram-matrix cone optimization with factor-side differentiable correction, and replacing ADMM with an unrolled spectral-proximal solver. More broadly, the paper points to a practical direction for geometric vision, i.e., modernize the solver while preserving the algebraic principle.

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